Limit comparison test — 5.4

$\sum_{n=1}^{\infty} \frac{n-1}{n^2}$ should have the same behavior

as $\sum_{n=1}^{\infty} \frac{1}{n}$ (divergence)

<u>Recall</u>: e^x grows faster than any polynomial

$$\lim_{x \to \infty} \frac{\text{polynomial}}{e^x} = 0$$

Limit Comparison Test: If $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ both have positive terms, then

(a) if $\lim_{n\to\infty} \frac{a_n}{b_n}$ exists and is nonzero, then these series either both converge or both diverge.

(b) if $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} b_n$ converges then also $\sum_{n=1}^{\infty} a_n$ converges.

(c) if $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum_{n=1}^{\infty} b_n$ diverges, then also $\sum_{n=1}^{\infty} a_n$ diverges.

If $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ for $L \neq 0$,

then $a_n \approx L \cdot b_n$ and $\sum_{n=1}^{\infty} a_n \approx L \cdot \sum_{n=1}^{\infty} b_n$

<u>Example</u>: $\sum_{n=1}^{\infty} \underbrace{\frac{n-1}{n^2}}_{a_n}$ is similar to $\sum_{n=1}^{\infty} \underbrace{\frac{1}{n}}_{b_n}$, and

we can now make this precise!

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{n-1}{n^2}}{\frac{1}{n}} = \lim_{n\to\infty} n \cdot \frac{n-1}{n^2} = \lim_{n\to\infty} \frac{n-1}{n} = 1$$

Thus by the LCT, the series both diverge since we know that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by the p-test.

Example: Does $\displaystyle\sum_{n=1}^{\infty} \frac{n^3+2n+100}{3n^6+10n^3-12}$ converge or diverge?

Want to compare to $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$ with the LCT.

$$\lim_{n\to\infty} \frac{\dfrac{n^3+2n+100}{3n^6+10n^3-12}}{\dfrac{1}{n^3}} = \lim_{n\to\infty} n^3 \cdot \frac{n^3+2n+100}{3n^6+10n^3-12}$$

$$= \lim_{n\to\infty} \frac{n^6+2n^4+100n^3}{3n^6+10n^3-12} = \frac{1}{3}$$ and this is nonzero,

so the LCT applies. Thus $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges

by the p-test implies that

$\displaystyle\sum_{n=1}^{\infty} \frac{n^3+2n+100}{3n^6+10n^3-12}$ also converges.

__Example__: Does $\sum_{n=1}^{\infty} \frac{2^n+1}{3^n}$ converge or diverge?

$\rightarrow$ looks similar to $\frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n$.

$$\lim_{n\to\infty} \frac{\frac{2^n+1}{3^n}}{\frac{2^n}{3^n}} = \lim_{n\to\infty} \frac{\cancel{3^n}}{2^n} \cdot \frac{2^n+1}{\cancel{3^n}} = \lim_{n\to\infty} \frac{2^n+1}{2^n}$$

$$= \lim_{n\to\infty} 1 + \frac{1}{2^n} = 1.$$

We know $\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ converges because it is a geometric series with common ratio less than 1, and thus the LCT implies that $\sum_{n=1}^{\infty} \frac{2^n+1}{3^n}$ also converges.